Arduino MEGA Multi-Sensor Monitoring and Control System

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RULE BEFORE STARTING

Do NOT connect everything at once.
You will build like this:

1. **Test I2C ONLY (no sensors)**
2. **Add devices ONE BY ONE**

MEGA-1: USER INTERFACE AND CONTROL SYSTEM

Overview

MEGA-1 is responsible for all user interactions. It handles passcode verification, temperature input, system start control, and sends the preferred temperature to MEGA-2 using I2C communication.

STEP 1: Power and Setup

Connect MEGA-1 to your computer using USB.
Upload the MEGA-1 code to the board.

After uploading, the board stores the program and will run automatically when powered.

STEP 2: Connect LCD (Display Unit)

LCD Pin Connections

LCD Pin	Arduino MEGA-1 Pin
RS	12
E	11
D4	5
D5	4
D6	3
D7	2

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LCD Pin	Arduino MEGA-1 Pin
VSS	GND
VDD	5V
RW	GND
VO	Potentiometer
A	5V (via resistor)
K	GND

Potentiometer Connection

- One side → 5V
- One side → GND
- Middle → VO (LCD contrast)

Adjust the potentiometer until text is visible on the LCD.

STEP 3: Connect Keypad (Passcode Input)

Keypad Pin Connections

Keypad Pin	Arduino Pin
R1	29
R2	28
R3	27
R4	26
C1	25
C2	24
C3	23
C4	22

The keypad is used to:

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- Enter passcode
- Confirm passcode
- Enter preferred temperature

STEP 4: Connect Push Button (PB1)

Connection

- One side → 5V
- Other side → pin 30
- Add 10kΩ resistor from pin 30 → GND (pull-down)

Purpose

PB1 is used to start the system after temperature is set.

STEP 5: Connect Active Buzzer

Connection

- Positive → pin 31
- Negative → GND

Purpose

The buzzer gives a confirmation sound when the passcode is successfully verified.

STEP 6: Connect 4-Digit 7-Segment Display (Using 74HC595)

74HC595 to Arduino

74HC595 Pin	Arduino Pin
DS	32
SH_CP	33
ST_CP	34

Digit Control Pins

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Digit	Arduino Pin
D1	35
D2	36
D3	37
D4	38

Important Control Pins

- MR → 5V
- OE → GND

Purpose

- Displays passcode
- Flashes passcode for 5 seconds after successful verification

STEP 7: Now build MEGA-2 hardware (slowly)

From here: you build ONE component at a time

STEP 7A: LCD FIRST (visual output)

Connect:

LCD Pin	Arduino
RS	22
E	23
D4	24
D5	25
D6	26
D7	27
VSS	GND

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LCD Pin	Arduino
VDD	5V
RW	GND
VO	potentiometer middle
A	5V (via resistor)
K	GND

Potentiometer:

- one side → 5V
- one side → GND
- middle → VO

Turn knob until text appears

STEP 7B: DHT11 (temperature sensor)

Connect:

- DATA → **28**
- VCC → 5V
- GND → GND

If raw sensor → add 10k resistor DATA → 5V

Test:

- LCD should show temp & humidity
- If "DHT Error" → wiring issue

STEP 7C: PIR + Yellow LEDs (motion)

PIR:

- OUT → **29**
- VCC → 5V
- GND → GND

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Yellow LEDs:

Pins:

- 48, 49, 50, 51

Each LED:

- pin → resistor → LED → GND

Test:

- move hand → all yellow LEDs ON

STEP 7D: Ultrasonic + Buzzer (safety)

Ultrasonic:

- TRIG → 30
- ECHO → 31
- VCC → 5V
- GND → GND

Buzzer:

- → 43
- → GND

Test:

- hand < 20 cm → buzzer ON

STEP 7E: LDR + Red LEDs (Night mode)

LDR:

- one side → 5V
- other → A0
- resistor (10k) A0 → GND

Red LEDs:

Pins:

- 44, 45, 46, 47

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Test:

- cover LDR → red LEDs ON

STEP 8: Motor (LAST STEP)

NEVER directly connect motor to Arduino

Correct way:

- pin **2 → transistor → motor**
- motor powered separately
- diode across motor

Logic:

- motion + high temp → motor ON
- object close → motor OFF

FINAL SYSTEM FLOW

When everything works:

1. MEGA-1 sends temperature
2. MEGA-2 receives it
3. PIR detects motion → yellow LEDs ON
4. DHT reads temp
5. If temp > preferred → motor ON
6. If object close → motor OFF + buzzer
7. If dark → red LEDs ON + temp forced to 22°C

STEP 9: I2C Connection to MEGA-2

Connection

MEGA-1	MEGA-2	Meaning
Pin 20	Pin 20	SDA (data line)
Pin 21	Pin 21	SCL (clock line)

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MEGA-1	MEGA-2	Meaning
GND	GND	Must

This is like:

- SDA = talking wire
- SCL = timing wire
- GND = common reference (without this NOTHING works)

Purpose

MEGA-1 sends the preferred temperature to MEGA-2.

STEP 10: System Logic Flow

Passcode Entry

1. User enters a 4-digit passcode using keypad
2. User re-enters the same passcode

If mismatch:

- Display "Mismatch"
- Restart process

If matched:

- Display "Matched"
- Activate buzzer for 1 second
- Flash passcode on 7-segment display for 5 seconds

Temperature Input

1. LCD prompts user to set temperature
2. User enters 3 digits (example: 235)
3. System converts:
    - 235 → 23.5°C

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- Also displays °F equivalent

System Start

1. LCD displays “Press PB1 to Start”
2. User presses PB1

Result:

- LCD shows “System is ON”
- Temperature value is sent to MEGA-2 using I2C

STEP 11: I2C Data Transmission

MEGA-1 sends temperature as a 3-digit number (example: 235).

Process

- Split into high byte and low byte
- Send both bytes via I2C

MEGA-2 receives and converts:

235 → 23.5°C

MEGA-1 RESPONSIBILITIES SUMMARY

- Handle keypad input
- Verify passcode
- Display messages on LCD
- Convert and display temperature (°C and °F)
- Control active buzzer
- Flash passcode on 7-segment display
- Detect PB1 press
- Send preferred temperature to MEGA-2 via I2C

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